# Progressive Spatial–Spectral Joint Network for Hyperspectral Image Reconstruction

Tianshuai Li and Yanfeng Gu, *Senior Member, IEEE*

***Abstract*— Hyperspectral (HS) images are widely used to identify and characterize objects in a scene of interest with high acquisition costs and low spatial resolution. It is an inexpensive way to obtain high-spatial-resolution HS images (HSIs) by spectral reconstruction from high-spatial-resolution multispectral (MS) images. In this article, we proposed a progressive spatial–spectral joint network (PSJN) to reconstruct HSIs from MS images. PSJN is composed of a 2-D spatial feature extraction module, a 3-D progressive spatial–spectral feature construction module, and a spectral postprocessing module. PSJN makes full use of the shallow spatial features extracted by the 2-D spatial feature extraction module with the spatial–spectral features extracted by the 3-D progressive spatial–spectral feature construction module. The 3-D progressive spatial–spectral feature construction module is designed to extract spatial–spectral information from local spectra in local space and construct spectral information from a few bands to a lot of bands in a pyramidal structure. Besides, a network updating mechanism is proposed to improve the spectral reconstruction effect of the images with poor original spectral reconstruction effect. The experimental results on three HS–MS datasets and one MS dataset demonstrate the efficacy of the proposed methods. Compared with the most advanced spectral reconstruction methods based on dictionary learning and deep learning, our method achieves the best performance of the latest methods in similarity evaluation and classification performance evaluation.**

***Index Terms*— Hyperspectral (HS) data, multispectral (MS) data, neural networks, spectral reconstruction (SR), spectral super-resolution.**

## I. Introduction

HYPERSPECTRAL images (HSIs) provide rich and diverse spectral information from a scene with hundreds or thousands of narrow spectral bands. Rich spectral characters contribute to distinguishing different objects. Consequently, HSIs are widely applied in classification [1], [2], target detection [3], intrinsic image recovery [4], change detection [5], spectral unmixing [6], and scene segmentation [7]. The single spectral band has low energy due to the narrow spectral band. Thus, hyperspectral (HS) sensors need to enlarge the instantaneous field of view to obtain a reasonable signal-to-noise ratio leading to spatial resolution limitation. In addition, space-borne HS data have a long revisit time due to the very limited HS satellites. By contrast, multispectral (MS) satellites have a higher spatial resolution (mostly lower than 10 m) and shorter revisit time by the satellite network. Some scholars try to add HSIs of different phases or different sensors into the used dataset through the transfer learning method [8]. However, available HS data are still too scarce.

The researchers try to set one resolution and improve another to solve the contradiction between the various resolutions in multimodal HS terms (essentially the contradiction between spectral resolution and spatial resolution) [9]. Initially, the researchers focused on improving the spatial resolution of HSIs. Many effects have been made to restore images with both high spectral resolution and spatial resolution by fusing high-spatial-resolution MS images with low-spatial-resolution HSIs or other relative information [10]–[13]. However, with the long reexamination period for HS satellites, it is challenging to collect well-recorded HSI and MSI at any given time and to meet our needs.

In view of the advantages of MS images, such as short revisit time and high spatial resolution, restoring the lost spectral information of MS images is also an effective method to resolve the contradiction between spectral resolution and spatial resolution, defined as spectral reconstruction or spectral super-resolution. Spectral reconstruction builds mapping from a few bands (three or 4–20) to a large number of bands ($>100$). This inverse process is a severely ill-posed problem. Many spectral reconstruction approaches were proposed to resolve the problem.

The early researchers adopt shallow mapping based on principle component analysis (PCA) or pseudoinverse (PI) [14]–[19]. The mapping method based on sparse dictionary learning emerged because of more prior knowledge [20]–[27]. With the development of deep learning, methods based on CNN or GAN were used in spectral reconstruction [28]–[37]. The powerful ability to express the feature and construct mapping relation of deep learning approaches has achieved considerable success in spectral reconstruction.

Nevertheless, there are still several shortcomings in the existing spectral reconstruction models based on deep learning methods, such as CNN or GAN. On the one hand, most of the existing spectral reconstruction models, which adopt a lot of upsampling, downsampling, and nonlocal attention structure, are applied to ground images. Due to the large-scale,

Manuscript received December 3, 2020; revised March 15, 2021 and April 14, 2021; accepted May 10, 2021. Date of publication May 24, 2021; date of current version January 5, 2022. This work was supported in part by the National Natural Science Foundation of Key International Cooperation under Grant 61720106002 and in part by the National Natural Science Foundation for Outstanding Scholars under Grant 62025107. *(Corresponding author: Yanfeng Gu.)*

The authors are with the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150001, China, and also with the Heilongjiang Province Key Laboratory of Space-Air-Ground Integrated Intelligent Remote Sensing, Harbin 150001, China (e-mail: guyf@hit.edu.cn).

Digital Object Identifier 10.1109/TGRS.2021.3079969

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

numerous, and complex ground objects of remote sensing images, these structures are difficult to play an excellent effect in the spectral reconstruction of remote sensing images. On the other hand, the spectral resolution of remote sensing HSI is high with a relatively low spatial resolution. The spectral correlation between bands in remote sensing images is stronger than the spatial correlation, and the continuity of adjacent spectra should be paid more attention to. However, existing models based on 2-D-CNN are difficult to express spectral continuity of adjacent spectra.

To address these issues, we propose a progressive spatial–spectral joint network (PSJN) combining spatial and spectral features of 2-D-CNN and 3-D-CNN. PSJN consists of a spatial–spectral feature construction module and a spectral postprocessing module. The spatial–spectral features construction module combines a 2-D spatial feature extraction module and a 3-D progressive spatial–spectral feature construction module. The progressive spatial–spectral feature extraction modules that make up the 3-D progressive spatial–spectral feature construction module generate high and low-frequency spectral information layer by layer. HSIs can be reconstructed from MS images with high accuracy through the above structures.

The main contributions of this article are summarized as follows.

1) A PSJN combining spatial features of 2-D-CNN and spatial–spectral features of 3-D-CNN is proposed for spectral reconstruction. The 2-D network extracts shallow spatial information of MS images. The 3-D network is used for the progressive construction of local spatial–spectral-related features. HSIs can be reconstructed accurately by combining spatial and spectral features.

2) A progressive pyramid structure at spectral dimension is proposed to recover the spectral information from low frequency to high frequency step by step through multilayer superposition. The 3-D convolution combines different deep features of the same spatial position and spectrum in 3-D progressive modules of the pyramidal structure.

3) A network updating mechanism is proposed to improve the spectral reconstruction effect of the images with poor original spectral reconstruction effect. We evaluate the quality of SR methods from similarity and classification performance evaluation in three HS–MS datasets and one MS dataset. Extensive experiments show the superiority of the PSJN model for HS reconstruction.

The remaining part of this article is organized as follows. Section II describes the development of spectral reconstruction methods. We present the framework of the proposed PSJN in Section III. The results and analysis of extensive experiences are presented in Section IV. Section V summarizes the conclusions of this work.

## II. Related Works

The early researchers expect to find a mapping matrix that can directly express the correlation between MS image and HSI. The typical methods are spectral reconstruction based on the principal component analysis (PCA) [14], [15], spectral reconstruction based on Wiener estimation (WEN) [16], and spectral reconstruction based on pseudoinverse (PI) [17], [18]. Over the past five years, spectral reconstruction methods have been divided into two branches: knowledge-driven spectral reconstruction methods based on sparse dictionary learning and data-driven spectral reconstruction methods based on deep neural network learning.

Spectral reconstruction methods based on the sparse dictionary expect to express MS image and HSI through the same weight matrix and different dictionaries. The corresponding HSIs can be obtained using the HS dictionary and the weight matrix obtained from the MS dictionary and images. Researchers have given many deformation methods through different prior information based on this basic idea.

Arad and Ben-Shahar [20] obtained an overcomplete dictionary from the HSIs by the K-SVD algorithm and obtained the lower dimension dictionary by projecting. Robles-Kelly [21] processes convolutional features through dictionary learning. Wu *et al.* [22] improved ARAD's sparse dictionary method by introducing an "A+"-based method. Jia *et al.* [23] proposed a two-step mapping method that used manifolds as the intermediate structure. Han *et al.* [24] constructed a sparse dictionary using the spectral library as auxiliary information. Yi *et al.* [25] reconstructed the spectrum by combining the sparse dictionary and unmixing models. Gao *et al.* [27] learned low-rank sparse dictionaries on the HS and MS images, respectively.

Spectral reconstruction methods based on deep learning hope to fit the inverse mapping by constructing a suitable deep neural network. Nguyen *et al.* [28] tried to fit the inverse mapping by a narrow RBF network. The CNN-based spectral reconstruction appeared later in 2017. Galliani *et al.* [29] first introduced a CNN mapping from low-dimension images to high-dimension images. Xiong *et al.* [30] proposed HSCNN that outputs the reconstructed HSIs with the input of the spectrally upsampled images. Alvarez-Gila *et al.* [31] adopted a generative adversarial network to reconstruct the spectrum. Paul and Kumar [32] proposed a spectral reconstruction idea that extracted features of a region to construct the spectrum of the central point and apply it to remote sensing images. Similar to spectral reconstruction, related deep learning techniques are also used in compressed sensing [38].

In 2018, Arad *et al.* [39] held the first spectral image reconstruction challenge named NTIRE 2018 Spectral Reconstruction Challenge, in which all participants proposed methods of deep learning. Shi *et al.* [33] developed a deep Resnet structure and a deep Densenet structure named HCSNN-R and HSCNN-D, respectively, and won the first space of the challenge. The second challenge on spectral reconstruction was held in 2020 [40]. All participating teams are using deep learning methods in NTIRE 2020 as well as what happened in 2018. Li *et al.* [34] obtained the first ranking on the "Clean" track with their adaptive weighted attention network (AWAN), and Zhao *et al.* [35] obtained the first ranking on the "Real World" track with their Hierarchical Regression Network (HRNet) [35].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

## III. Proposed Method

### A. Problem Formulation

Both HS and MS images are subsamples of continuous spectral images. HSIs retain more information than MS images, which loses much spectral information during sampling by MS sensors. The observed value of MS sensors on each band is related to the value of continuous spectrum and spectral response function (SRF) of MS sensors. Let $L$ and $L_M$ be separately a continuous signal and an MS signal. The relationship between $L$ and $L_M$ can be described by the SRF of the MS sensor $g$

$$L_M(i) = \frac{\int L(\lambda) g_i(\lambda) d\lambda}{\int g_i(\lambda) d\lambda} \tag{1}$$

where $i$ is designated band of $L_M$, $g_i$ is the SRF of the MS sensor on band $i$, $\lambda$ is the wavelength, and the value range of $\lambda$ depends on sensor and $i$.

Since $\int g_i(\lambda) d\lambda$ is a fixed value for the specified band of the specified sensor, we can describe (1) by the normalized SRF

$$L_M(i) = \int L(\lambda) \frac{g_i(\lambda)}{\int g_i(\lambda) d\lambda} d\lambda = \int L(\lambda) \overline{g}_i(\lambda) d\lambda. \tag{2}$$

In fact, the integral formula above is generally approached in discrete shapes. Therefore, (2) can also be written as follows:

$$L_M(i) = \sum_{n=1}^{N} L(n) \overline{g}_i(n) \tag{3}$$

where $N$ is the number of points sampled by the SRF of the MS sensor.

For example, SRF of GF1 covers the band from 400 to 1000 nm and $N$ of GF1 values for 601. The observed value of each band of GF1 can be considered the convolution samples of the actual continuous spectral radiance on 601 bands with the SRF of length 601.

Due to the high spectral resolution of HSIs and the narrow bandwidth of HS sensor SRF, we can interpolate the HS spectrum to get the continuous spectrum. On this basis, MS images can be obtained in continuous spectrum images derived from HSI interpolation with the help of MS sensor SRF [41].

Therefore, the SRF of the specified bands from HSIs to MS images $\overline{g_{\mathrm{HM}}}$ can be obtained by interpolating the weight of HS bands and SRF of MS sensors

$$L_M(i) = \sum_{n=1}^{B_H} L_H(n) \overline{g_{\mathrm{HM}i}}(n) \tag{4}$$

where $L_H$ is the HS signal and $B_H$ denotes the number of bands.

Therefore, (4) is equivalent to

$$L_M = G_{\mathrm{HM}} L_H. \tag{5}$$

SRF of GF1 and normalized SRF of specified bands from GF5 to GF1 are shown in Fig. 1.

Spectral reconstruction from MS images to HSIs is actually an inverse mapping of (5). $G_{\mathrm{HM}}$ has no left inverse matrix because it is a row full rank matrix. There is an infinite number of solutions to $L_H$ when $L_M$ is fixed. Therefore, it is impossible to directly find an inverse mapping matrix according to the MS sensor's SRF to complete spectral reconstruction. Reverse mapping itself is intrinsically ill-placed. Arad and Ben-Shahar [20] demonstrated the feasibility of ill-posed transformation for HS signal's sparsity in natural scenes.

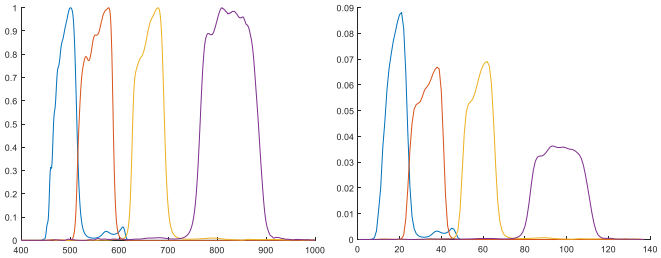


Fig. 1. SRF of GF1 and normalized SRF of 137 bands from GF5 to GF1.

Spatial continuity and spectral continuity exist between local pixels and local bands in HSIs of natural scenes. Reasonable utilization of space continuity and spectral continuity helps to realize spectral reconstruction from MS images to HSIs.

However, reconstruction may fail if the data distribution of test MS images utterly inconsistent with the data distribution of MS images in the training set is encountered. In this case, updating mechanism can be used for spectral reconstruction.

Based on the above analysis, the overall workflow of this article is shown in Fig. 2. First, HS–MS sample pairs are constructed by overlapping HSIs and MS images. Second, the designed network PSJN is used to reconstruct MS samples. The loss is calculated according to the difference between reconstructed HS samples and real HS samples to constrain the network to train in a more accurate direction. Finally, perform spectral reconstruction from MS images to HSIs by the trained network.

In the process of spectral reconstruction, we can self-evaluate the reconstruction effect of HSIs and carry out network updating on samples with a poor reconstruction effect, so as to achieve a better spectral reconstruction effect.

### B. Network Architecture

As shown in Fig. 3, our proposed PSJN consists of three parts: the 2-D spatial feature extraction module (2-D SFEM), the 3-D progressive spatial–spectral feature construction module (3-D PSFCM), and the spectral postprocessing module.

Transmitted features have four dimensions in the 2-D spatial feature extraction module: a quantity dimension, a spectral dimension (channel dimension of 2-D convolution), and two spatial dimensions.

Transmitted features have four dimensions in the 3-D progressive spatial–spectral feature construction module: a quantity dimension, a feature dimension (channel dimension of 2-D convolution), a spectral dimension, and two spatial dimensions.

Given $I_M \in N \times M \times X \times Y$ ($N$ is the number of samples, $M$ is the number of MS bands, and $X$ $YX$ and $Y$ are the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

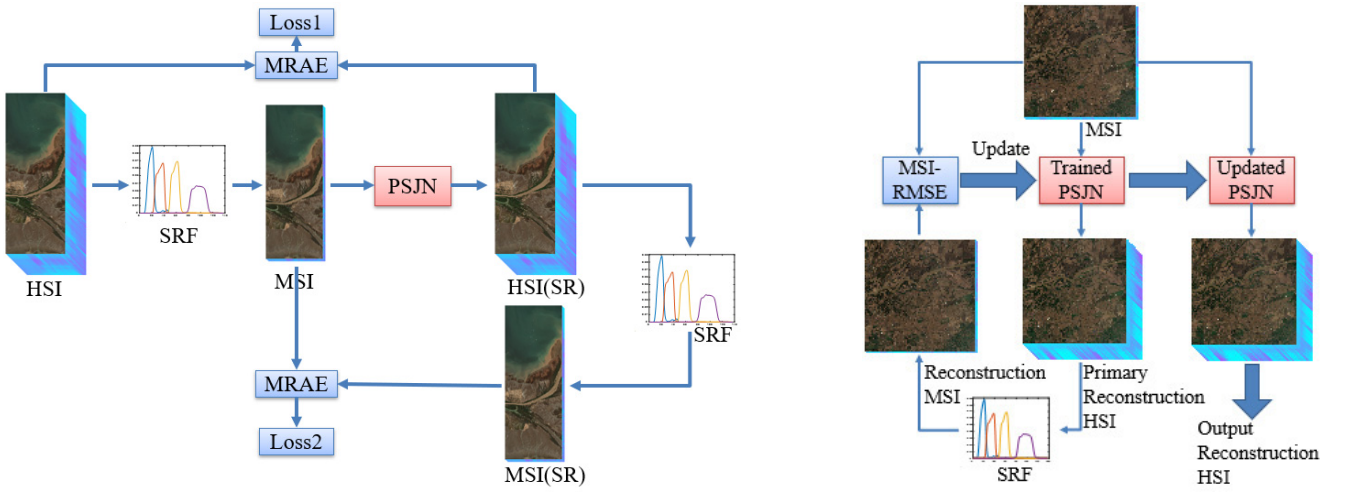



Fig. 2. Workflow of the proposed PSJN. (Left) Train. (Right) Test.

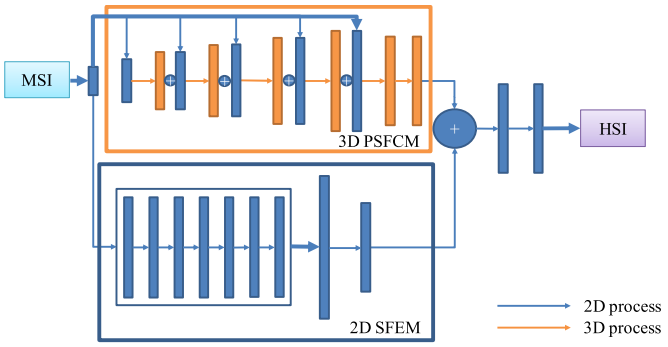



Fig. 3. Network architecture of our PSJN.

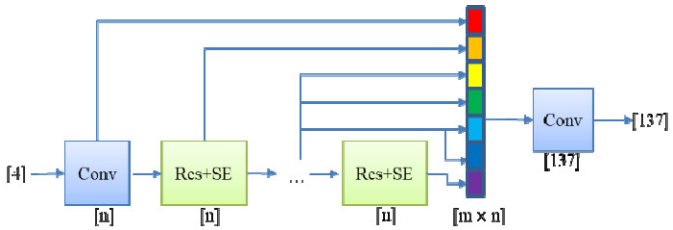



Fig. 4. Structure of 2-D SFEM. Letters in brackets denote the dimensionality of the spectral dimension.

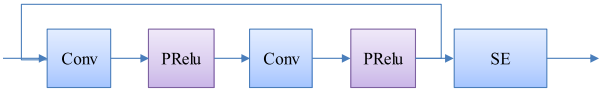



Fig. 5. Residual channel attention module.

dimensionalities of spatial dimensions) as the MS input. The output of the 2-D spatial feature extraction module is

$$I_{\text{2-D}} = H_{\text{2-D SFEM}}(I_M) \tag{6}$$

where $H_{\text{2-D SFEM}}$ represents system response of the 2-D spatial feature extraction module, $I_{\text{2-D}} \in N \times H \times X \times Y$ ($H$ is the number of HS bands).

The output of another channel 3-D PSFCM in the network is

$$I_{\text{3-D}} = H_{\text{3-D PSFCM}}(I_M) \tag{7}$$

where $H_{\text{3-D PSFCM}}$ represents system response of the 3-D progressive spatial–spectral feature construction module, $I_{\text{3-D}} \in N \times 1 \times H \times X \times Y$ (1 is the feature dimension).

After squeezing the extra second dimension, the output becomes $I_{\text{3-Dsq}} \in N \times H \times X \times Y$.

The spectral postprocessing module's input is the summation of the output of 2-D SFEM and 3-D PSFCM. The output of the spectral postprocessing module is the restored HS sample

$$I_{\text{SR}} = H_{\text{SE}}(H_{\text{2-D SFEM}}(I_M) + H_{\text{3-D PSFCM}}(I_M)) = H_{\text{PSJN}}(I_M) \tag{8}$$

where $H_{\text{SE}}$ represents the system response of the spectral postprocessing module.

### C. 2-D Spatial Feature Extraction Module

Fig. 4 shows that 2-D SFEM comprises three parts: the dimensionality ascending layer, the feature processing layer, and the dimensionality decreasing layer.

The dimensionality ascending layer consists of a 2-D convolution layer and an activation layer used to raise the number of data channels from the number of original MS samples to the specified spectral dimensionality (256 is used in the experiment).

The feature processing layer consists of several residual channel attention modules and a feature concatenate module.

As shown in Fig. 5, the residual channel attention module consists of a residual block [42] and a channel attention block. Two 2-D convolution layers and two parametric rectified linear unit (PRelu) layers [43] alternate in the residual block.

The channel attention block adopts the squeeze-and-excitation (SE) structure proposed by [44]. As shown in Fig. 6, the SE block comprises a global average pooling layer, two fully connected layers, an Relu layer, and a sigmoid layer.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

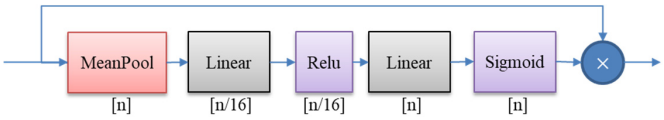



Fig. 6. SE module.

The SE block improves representations produced by explicitly modeling the interdependencies between the channels of features.

The 2-D spatial feature extraction module has sufficient depth to fully extract the features by stacking the residual channel attention module. Feature maps from the dimensionality ascending layer and channel attention blocks are combined in the channel dimension to combine features from different depths.

Dimensionality decreasing layer consists of a 2-D convolution layer and an activation layer, which is used to reduce the number of data channels to the number of real HS bands.

### *D. 3-D Progressive Spatial–Spectral Feature Construction Module*

The 3-D PSFCM comprises several progressive spatial–spectral feature extraction modules (progressive module for short) and a dimensionality decreasing layer at the end. The structure of the progressive module is shown in Fig. 7.

There are two inputs and one output in the progressive module. The inputs are the original MS sample in four dimensions and the output of the upper progressive module in five dimensions. The output is 5-D data with a multiplicative spectral dimension.

The progressive module consists of five parts: the dimensionality ascending layer, the feature extraction layer, concatenation in feature dimension, the summation layer, and the 3-D transposed convolution layer.

Each progressive module has its specified spectral dimensionality, which starts with a dimensionality slightly higher than the number of MS bands and multiplies step by step until the last progressive module exceeds the number of HS bands. The dimensionality ascending layer is used to raise spectral dimensionality from the number of MS bands to the specified dimensionality, consisting of a convolution layer and an activation layer.

Similar to the 2-D module, the feature extraction layer consists of several residual attention modules used to extract features at different depths. It should be noted that the features extracted by the different progressive blocks are also based on their respective specified spectral dimensionality. In addition, constrained by the subsequent 3-D convolution and 3-D transposed convolution, the order for the spectral dimension of the acquired features themselves is directly related to the real spectrum order.

After feature extracting, concatenation is used to combine feature maps of different depths. In contrast to 2-D SFEM, which combines feature maps in the spectral dimension, 3-D PSFCM combines feature maps in the feature dimension (added dimension). After the combination, the dimensionality of the spectral dimension remains the same. Meanwhile, the dimension of the data has changed from four to five. The advantage is that the previously extracted feature maps are constrained to be relevant at specified points (the same spatial dimension and the same spectral dimension). As shown in Fig. 8, the output of the 3-D convolution is only correlated with the adjacent spectral features and without the distant spectral features by stacking features in the feature dimension. The spectral continuity is constrained by local processing in the spectral dimension.

Data after concatenation and output of upper progressive module combine by direct addition with the same dimensionality.

The last part of the progressive module is the transposed convolution layer. The addition obtained in the previous step is put into the 3-D transposed convolution layer (stride is 2 in spectral dimension) to obtain the output of spectral dimensionality doubling. The dimensionality of the feature dimension of input and output depends on the number of residual attention modules in the upper and lower progressive modules.

The spectral dimension's dimensionality is increased to exceed the number of HS bands through several progressive blocks. The final progressive block no longer contains the transposed convolution layer but a 3-D convolution layer that reduces the spectral dimensionality to the same number of HS bands.

### *E. Loss Function and Others*

The reconstructed HSI should have two characteristics.

One is that the reconstructed HSI ought to be similar enough to the real HSI. The other is that the reconstructed HSI can be restored to the original MS image through the MS sensor's SRF.

Based on these two characteristics, the loss function of the PSJN network contains two parts: difference measurement between reconstructed HSIs and real HSIs, and difference measurement between reconstructed MS images and real MS images

$$l_H = \mathrm{MRAE}(I_H - H_{\mathrm{PSJN}}(I_M)) \tag{9}$$

$$l_M = \mathrm{MRAE}(I_M - H_{\mathrm{PSJN}}(I_M)\mathrm{SRF}_M). \tag{10}$$

The definition of mean relative absolute error (MRAE) is shown in Section IV.

Two loss functions are combined by linear addition

$$l = l_H + l_M. \tag{11}$$

The spectral reconstruction effect is improved during training by considering the difference measurement between reconstructing MS images and real MS images. However, considering that the distribution of training samples may be inconsistent with the distribution of real MS images, reconstructed MS images may still be inconsistent with the test samples' MS images. In this case, the difference can be used as an indirect measure to adjust the network.

Specifically, we designed two updating evaluation metrics for each sample block. One is the root mean square

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

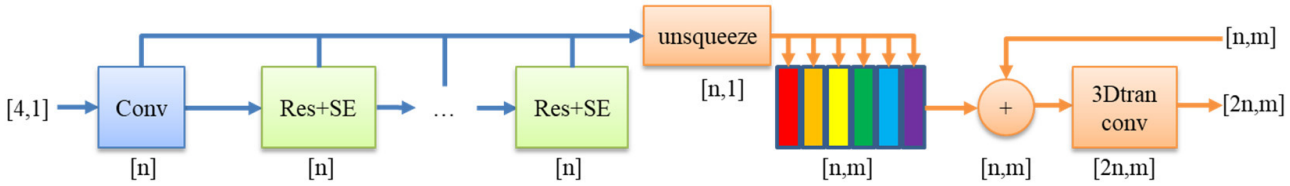



Fig. 7. Structure of the progressive spatial–spectral feature extract module. Letters in brackets denote the dimensionality of the spectral dimension and feature dimension.

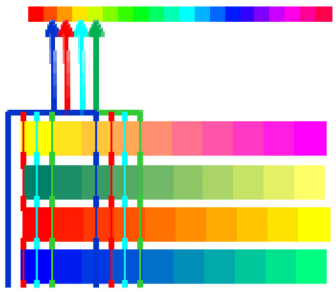


Fig. 8. 3-D convolution on spectral dimension.

TABLE I
DIMENSIONALITY SETTING OF THE SPECTRAL DIMENSION FOR LAYERS OF THE PSJN AND NPSJN

| Method | 2D SFEM | 3D PSFCM | Post-SE |
|---|---|---|---|
| PSJN | 4-256-256-1792-137 | 4-16-32-64-128-256-200-137 | 137-137 |
| NPSJN | 4-256-256-1792-137 | 4-9-18-36-72-144-137 | 137-137 |

error (RMSE) between the MS sample and the reconstructed MS sample, and the other is the number of pixels whose absolute difference value between the MS pixel and the reconstructed MS pixel exceeds a certain threshold. If one of them exceeds the limit, it is considered that the difference in distribution is too large and needs to be adjusted.

For blocks with a massive difference in distribution, the current MS sample is used as the training sample, and RMSE is used as the loss to carry out retraining. After 20 epochs of training or reaching the evaluation metrics, the training process stops, and the current retraining network reconstructs the MS samples.

The dimensionality setting of the spectral dimension for each layer of the PSJN structure is listed in Table I.

Considering a large amount of computing time brought by 3-D convolution, we also proposed a simplified version, called narrow PSJN (NPSJN). NPSJN reduces the spectral dimension of the 3-D PSFCM as a whole and reduces the amount of computation brought by 3-D convolution, thus reducing the operating time.

## IV. EXPERIMENTS

We evaluate the quality of SR methods from similarity and classification. In addition, we select four baselines, including SR based on the PI mapping matrix [17], SR based on sparse dictionary (ARAD [20] and J-SLoL [27]), and SR based on deep learning (AWAN [34]) in comparison with our model PSJN and NPSJN.

### A. Dataset Description

We have used image scenes from GF5 AHSI, AVIRIS, and GF1 PMS (see Table II) as HS and MS data. GF5 AHSI has 330 spectral bands, a 60-km swath width, and a 30-m spatial resolution. One hundred thirty-seven bands from 400 to 1000 nm with no band noise were selected as HS bands in the experiment. AVIRIS has 220 spectral channels. GF1 PMS has four spectral bands, a 30-km swath width, and an 8-m spatial resolution. We use the SRF of GF1 PMS to obtain the corresponding MS images of HSIs.

*1) GF5 Yellow River Delta Reserve Data (0531 for Short):* The first HS data were collected over the Yellow River Delta Reserve, China (E119.4, N37.7), by the GF5 satellite on May 31, 2019, as shown in Fig. 9(a). The local area with the land as the main part (1101 × 1050 pixels) was selected, in which the number of effective pixels is 972 928. A subimage is selected with a size of 1016 × 376 as the training area, as shown in Fig. 10(a).

The training area was divided into 22 500 overlapping sample blocks of 16 × 16 pixels as the deep learning method's training set. In addition, 768 sample blocks of 16 × 16 pixels were uniformly sampled from the full figure as the verification set. The main scenes in the figure were the sea, mudflat, and plants. There were seven classes in the studied scene: spartina, reed, tamarix, suaeda, mudflat, salt marsh, and water.

*2) GF5 Yellow River Estuary Town Data (0417 for Short):* The second HS data were collected over the Yellow River estuary town, China (E118.8, N37.7), by the GF5 satellite on April 17, 2019, as shown in Fig. 9(b). The entire region was taken as the testing set, among which the effective pixel number was 323 9710. The main scenes in the figure were sea, field, and buildings.

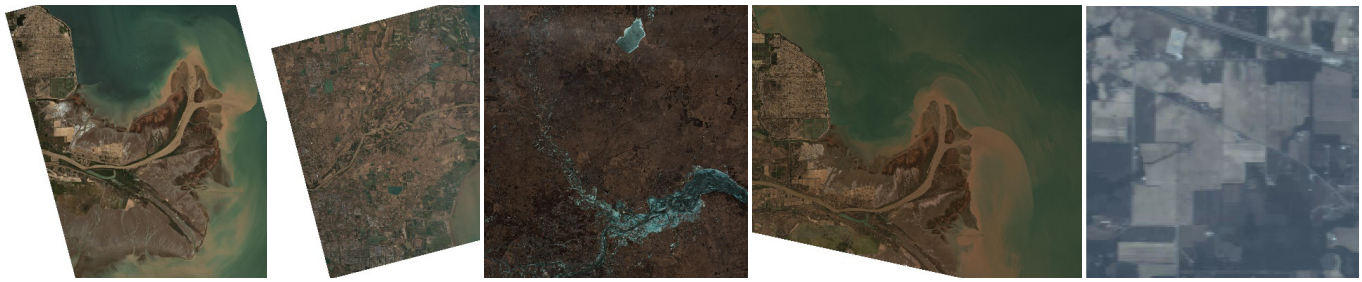
*3) GF5 Harbin Urban Area Data (0224 for Short):* The third HS data were collected over the Harbin urban area, China (E126.7, N46.1), by the GF5 satellite on February 24, 2019, as shown in Fig. 9(c). The entire region was taken as the testing set, among which the effective pixel number was 408 5056. The main scenes in the figure were river, ice, field, and buildings. There were seven classes in the studied scene: water, tree, bare land, field, village, residential, and high rise.

*4) GF1 Yellow River Delta Reserve Data (0506 for Short):* The first MS data were collected over the Yellow River Delta Reserve, China (E119.4, N37.7), by the GF1 satellite on May 6, 2019, as shown in Fig. 9(d). The main scenes in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SPECIFICATION OF SATELLITE DATA USED IN THIS STUDY

| Dataset Sign | Satellite | Sensor | Acquisition date | Acquisition location | Num of pixels | Spatial resolution | Application |
|---|---|---|---|---|---|---|---|
| 0531 | GF5 | AHSI | 31/05/2019 | Yellow River delta reserve E119.4_N37.7 | Train:0.38m Val:0.20m Test:0.59m | 30m | Model building and testing |
| 0417 | GF5 | AHSI | 17/04/2019 | Yellow River estuary town E118.8_N37.7 | Test:3.24m | 30m | Testing |
| 0224 | GF5 | AHSI | 24/02/2019 | Harbin urban area E126.7_N46.1 | Test:4.09m | 30m | Testing |
| 0506 | GF1 | PMS | 06/05/2019 | Yellow River delta reserve E119.3_N38.0 | Test:19.1m | 8m | Testing |
| IP | Airborne | AVIRIS | 12/06/1992 | Northwest Tippecanoe County, Indiana | Train:6525 Test:14500 | 20m | Model building and testing |



(a) (b) (c) (d) (e)

Fig. 9. Three used HS–MS datasets for evaluation of SR methods from similarity and classification and one MS datasets for evaluation of SR methods from classification. (a) GF5 Yellow River delta reserve (0531). (b) GF5 Yellow River estuary town (0417). (c) GF5 Harbin urban area (0224). (d) GF1 Yellow River delta reserve (0506). (e)AVIRIS IP.




(a) (b) (c)

Fig. 10. (a) Training set used for spectral reconstruction. (b) Training set used for classification on 0531. (c) Testing set used for classification on 0531.

figure were the sea, mudflat, and plants. There were seven classes in the studied scene: spartina, reed, tamarix, suaeda, mudflat, salt marsh, and water.

*5) AVIRIS Indian Pines (IP) Data:* This widely used HSI was collected over the Indiana state, USA, and consists of 145 × 145 pixels, as shown in Fig. 9(e). To make a relatively fair comparison, we adopt a set of training and test samples as used in research [27] (the corresponding MS image is simulated by using the SRFs of Sentinel-2 and a subimage is selected with the size of 145 × 45 as the partially overlapped region of HS and MS images). There were 16 classes in the studied scene.

The details about training and test sets for the three datasets 0531, 0224, and 0506 are shown in Table III. The training set test set of 0531 is a fixed set, as shown in Fig. 10(b) and (c). The training set and the test set of 0506 were selected to calibrate areas with basically the same geographical locations as the training set and test set of 0531.

TABLE III
DETAILS OF THE GROUND TRUTH IN DATASETS 0531, 0224, AND 0506

| 0531 | Train | Test | 0224 | Train | Test | 0506 | Train | Test |
|---|---|---|---|---|---|---|---|---|
| Spartina | 135 | 1204 | Water | 30 | 508 | Spartina | 1044 | 9497 |
| Reed | 46 | 381 | Tree | 30 | 90 | Reed | 382 | 3583 |
| Tamarix | 51 | 541 | Bare land | 30 | 52 | Tamarix | 546 | 5727 |
| Suaeda | 88 | 287 | Field | 30 | 585 | Suaeda | 658 | 1596 |
| Mudflat | 60 | 658 | Village | 30 | 403 | Mudflat | 468 | 6778 |
| Salt marsh | 16 | 89 | Residential | 30 | 409 | Saltmarsh | 72 | 540 |
| Water | 144 | 1989 | High-rise | 30 | 313 | Water | 885 | 13444 |
| Total | 540 | 5149 | Total | 240 | 2360 | Total | 4055 | 41165 |

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

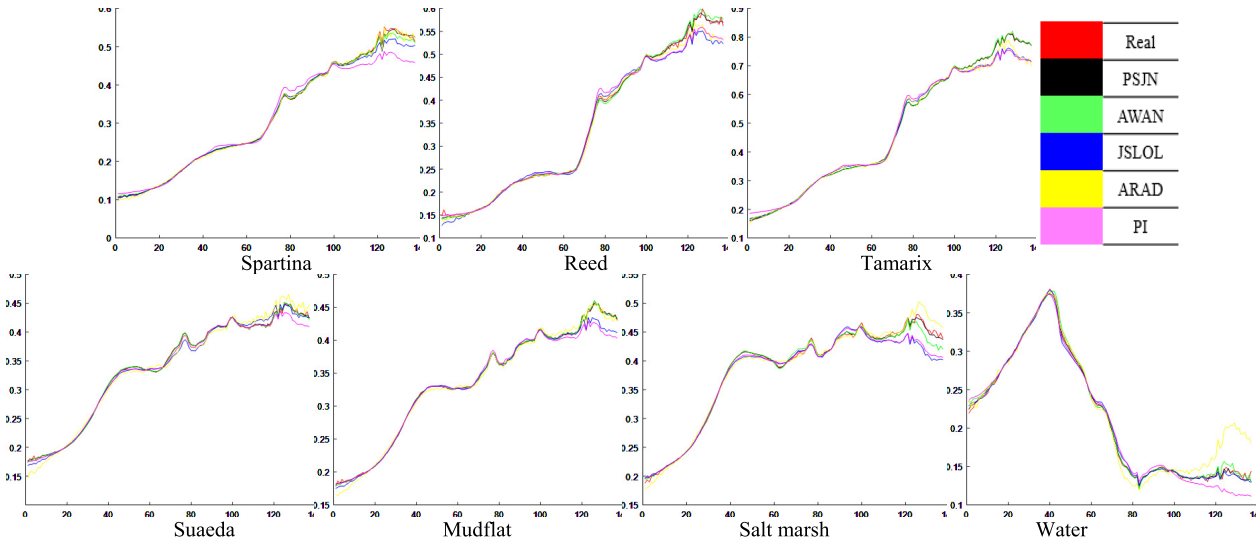



Fig. 11. Spectral reflectance curves for seven classes in dataset 0531. Red: real spectrum, black: PSJN, green: AWAN, blue: JSLOL, yellow: ARAD, and purple: PI.

Thirty samples are randomly selected from each class as the training set of 0224.

### B. Evaluation Metrics

For the HS–MS dataset, the reconstruction quality is measured by the similarity metrics between reconstructed HSIs and real HSIs. The specific evaluation metrics are RMSE, MRAE, and spectral angle mapper (SAM). The formulas of RMSE, MRAE, and SAM are given as follows:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_c \left(P_{gt_c} - P_{\mathrm{rec}_c}\right)^2}{N}} \tag{12}$$

$$\mathrm{MRAE} = \frac{\sum_c \frac{\left|P_{gt_c} - P_{\mathrm{rec}_c}\right|}{P_{gt_c}}}{N} \tag{13}$$

$$\mathrm{SAM} = \cos^{-1} \frac{P_{gt}^T P_{\mathrm{rec}}}{\left(P_{gt}^T P_{gt}\right)^{1/2} \left(P_{\mathrm{rec}}^T P_{\mathrm{rec}}\right)^{1/2}}. \tag{14}$$

For labeled datasets, the reconstruction index will be indirectly measured by the degree of improvement in classification ability.

The SVM classifier classified the training samples and test samples to evaluate the classification ability, as shown in Table III. The evaluation metrics are overall accuracy (OA), average accuracy (AA), kappa coefficient (KAPPA), and accuracy for each class.

### C. Similarity-Based Evaluation

The four test sets used for similarity metrics represent the four application environments encountered in practice.

The test set 0531 is consistent with training set 0531 at collection time, collection place, and distribution of ground objects. Dataset 0531 is used to verify the spectral reconstruction ability of spectral reconstruction methods on the nonoverlapping area of a group of partially overlapping HSI and MSI. The test set 0417 is slightly different from the training set in terms of collection time and distribution of ground objects. Dataset 0417 is used to verify the spectral reconstruction ability of spectral reconstruction methods on MS images of different phases. The test set 0224 is entirely different from the training set in terms of collection time, collection location, and distribution of ground objects and is used to verify the spectral reconstruction ability of spectral reconstruction methods on any MS images of the same sensor. Dataset IP is used to verify the spectral reconstruction ability of spectral reconstruction methods on a small training set.

TABLE IV
RECONSTRUCTION PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF THREE METRICS ON DATASET 0531. THE BEST RESULTS ARE SHOWN IN BOLD

| 0531 | RMSE | MRAE | SAM |
|---|---|---|---|
| PI | 0.02430 | 0.03860 | 0.06503 |
| ARAD | 0.01815 | 0.02552 | 0.04631 |
| J-SLoL | 0.01903 | 0.02050 | 0.04041 |
| AWAN | 0.00807 | 0.01089 | 0.02024 |
| PSJN | **0.00738** | **0.01005** | **0.01897** |
| NPSJN | **0.00744** | **0.01015** | **0.01919** |
| Ideal | ↓0 | ↓0 | ↓0 |

Tables IV–VII list the quantitative comparison between different spectral reconstruction algorithms in terms of RMSE, MRAE, and SAM as the similarity metrics on four HS–MS datasets 0531.0417, 0224, and IP.

The training and test sets' data distribution on dataset 0531 is consistent because the training and test samples are located in the adjacent area simultaneously.

Among the five methods, the PI method's similarity metrics are the worst and are much lower than other algorithms. It can be considered that it is still not feasible to use a single mapping matrix to directly represent the inverse mapping relationship from MS images to HSIs even under the circumstance that the data distribution of training and test sets are the same.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

TABLE V
RECONSTRUCTION PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF THREE METRICS ON DATASET 0417. THE BEST RESULTS ARE SHOWN IN BOLD

| 0417 | RMSE | MRAE | SAM |
|---|---|---|---|
| PI | 0.02075 | 0.02666 | 0.04853 |
| ARAD | 0.01748 | 0.02723 | 0.04600 |
| J-SLoL | 0.01677 | 0.02424 | 0.04103 |
| AWAN | 0.01467 | 0.01768 | 0.03309 |
| PSJN | **0.01366** | **0.01662** | **0.03175** |
| NPSJN | **0.01428** | **0.01740** | **0.03306** |
| Ideal | ↓0 | ↓0 | ↓0 |

TABLE VI
RECONSTRUCTION PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF THREE INDICES ON DATASET 0224. THE BEST RESULTS ARE SHOWN IN BOLD

| 0224 | RMSE | MRAE | SAM |
|---|---|---|---|
| PI | 0.02263 | 0.03755 | 0.06358 |
| ARAD | 0.01853 | 0.03075 | **0.05304** |
| J-SLoL | 0.01701 | 0.02966 | 0.05753 |
| AWAN | 0.01731 | 0.03205 | 0.05629 |
| PSJN | **0.01696** | **0.02823** | 0.05369 |
| NPSJN | **0.01642** | **0.02913** | 0.05403 |
| PSJN-UD | **0.01631** | **0.02753** | **0.05256** |
| Ideal | ↓0 | ↓0 | ↓0 |

TABLE VII
RECONSTRUCTION PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF THREE INDICES ON DATASET IP. THE BEST RESULTS ARE SHOWN IN BOLD

| IP | RMSE | MRAE | SAM |
|---|---|---|---|
| PI | 0.00466 | 0.01741 | 0.01309 |
| ARAD | 0.00572 | 0.02089 | 0.01668 |
| J-SLoL | 0.00387 | 0.01079 | 0.01162 |
| AWAN | 0.00435 | 0.01438 | 0.01237 |
| PSJN | **0.00345** | **0.00887** | **0.01032** |
| Ideal | ↓0 | ↓0 | ↓0 |

Among the remaining methods, it can be seen from the results of dataset 0531 that the reconstruction effect of the AWAN and our proposed PSJN method based on deep learning is far better than that of ARAD and J-SLoL based on dictionary learning. Under the condition that the data distributions of the training set and the test set are the same, the inverse mapping relationship learned by the deep learning methods' powerful fitting ability is extremely accurate so that MS images can be accurately reconstructed into HSIs. The PSJN method proposed by us is about 7% higher than the second-best method in various similarity metrics. In addition, although the reconstruction precision of NPSJN as a simplified version is somewhat lower than that of PSJN, it still has some advantages compared with other methods.

Real spectral reflectance curves and generated spectral reflectance curves for seven classes in dataset 0531 are compared in Fig. 11 (red: real spectrum, black: PSJN, green: AWAN, blue: JSLOL, yellow: ARAD, and purple: PI). In addition, we also attempt to compare the difference between reconstructed HSIs and real HSIs at a series of bands. The visual difference between the reconstructed image and the real image was simultaneously compared in eight bands that are uniformly selected from the blue band to the near-infrared band (475.9, 540.1, 604.3, 668.4, 732.5, 796.7, 860.9, and 925.0 nm) in Fig. 12. It can be seen that there is a tiny visual difference between the real HSIs and the reconstructed HSIs obtained by PJSN, which broadly proves the effectiveness of our method PSJN.

As the data distribution of dataset 0417 differs from the training set, the similarity metrics on the reconstructed HSI of various methods are reduced compared with dataset 0531. The conclusion obtained from dataset 0417 is still that the spectral reconstruction effect of deep learning methods is better than that of dictionary methods, and the spectral reconstruction effect of dictionary methods is better than that of the matrix method. However, the difference between methods in dataset 0417 is a little smaller than that in dataset 0531. Compared with other methods, the proposed PSJN and NPSJN methods have the optimal spectral reconstruction effect on the three metrics.

We show the spectral reconstruction effects of an abandoned reservoir and its surrounding farmland area in dataset 0417 by different methods in the form of the SAM diagram, as shown in Fig. 13. It can be seen that, in the bare soil in the middle area of the reservoir and the surrounding farmland area, our method has a better visual effect of spectral reconstruction than other methods.

As the data distribution of dataset 0224 is dramatically different from the training set, the reconstructed similarity metrics of various methods are reduced compared with dataset 0531. However, it can still be seen that the proposed PSJN and NPSJN methods have the optimal reconstruction effect compared with other methods. Considering the vast difference between the test set and the training set, we used the updating mechanism mentioned above to enhance the reconstruction effect. As shown in the PSJN-UD row in Table V, the reconstruction accuracy is roughly improved by 3%. The visual difference between the reconstructed image and the real image was simultaneously compared in eight bands that are uniformly selected from the blue band to the near-infrared band (475.9, 540.1, 604.3, 668.4, 732.5, 796.7, 860.9, and 925.0 nm) in Fig. 14.

The spectral reconstruction effect on the small dataset IP (thousand pixels) is different from other datasets (million pixels). The spectral reconstruction effect of PI is increased, while the effect of ARAD and AWAN is decreased due to insufficient data. The proposed PSJN is still applicable to small datasets and maintains an excellent spectral reconstruction effect.

Our method performs better than others at a comprehensive increase of around 0.00042 RMSE, 0.00192 MRAE, and 0.00130 SAM compared to the second-best method,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

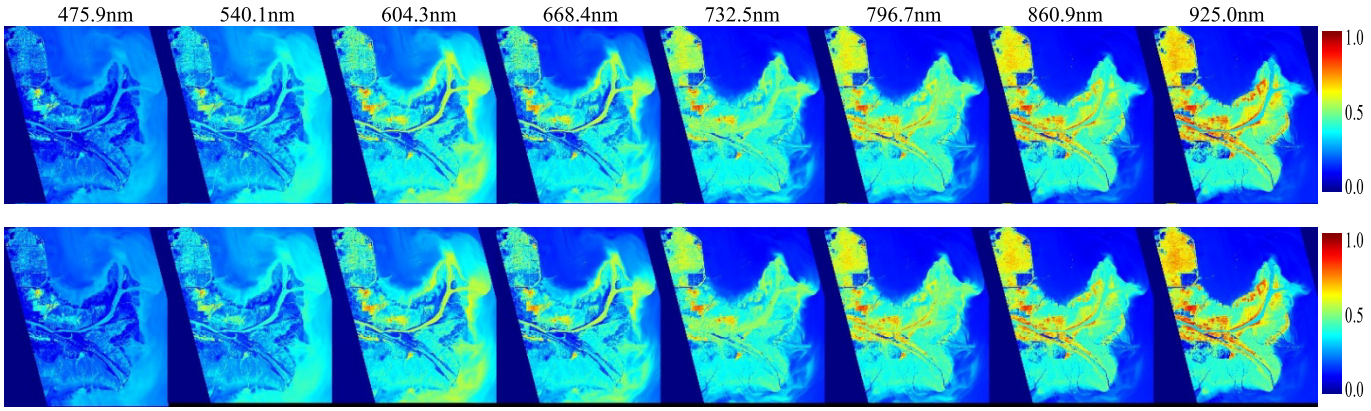



Fig. 12. Selective band visualization of the real HSI and the SR product using the proposed PSJN model on 0531 data. The first layer shows real bands, and the second layer shows spectral reconstruction bands.

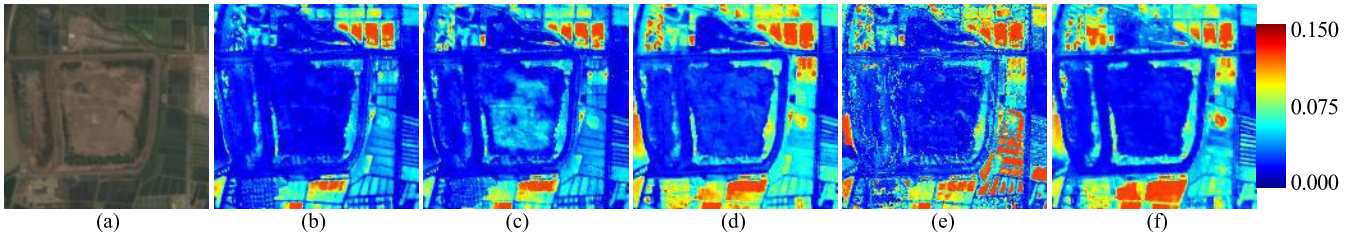



Fig. 13. Spectral reconstruction effects of an abandoned reservoir and its surrounding farmland area in dataset 0417 by different methods in the form of the SAM diagram. (a) Real HSI (RGB). (b) PSJN (SAM). (c) AWAN (SAM). (d) J-SLoL (SAM). (e) ARAD (SAM). (f) PI (SAM).

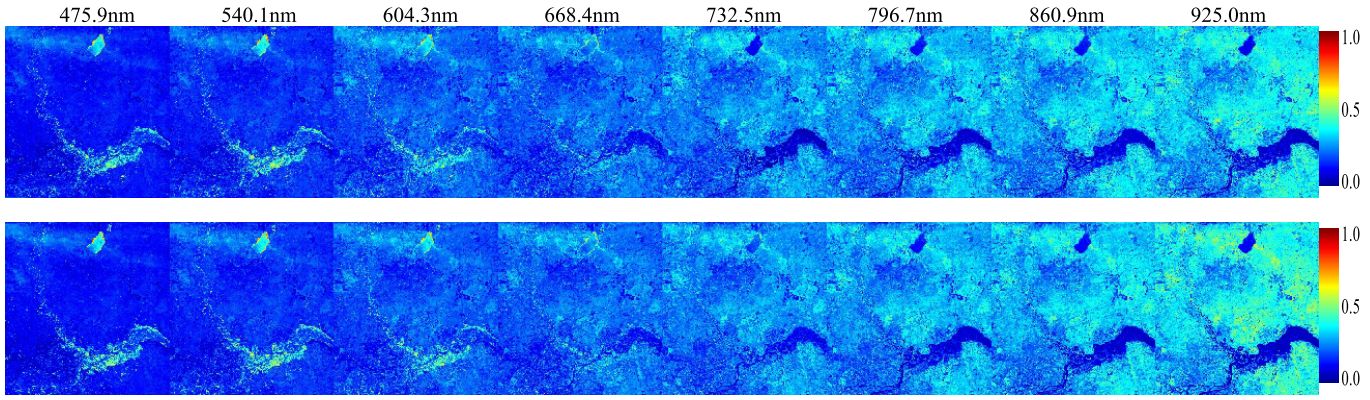



Fig. 14. Selective band visualization of the real HSI and the SR product using the proposed PSJN model on the 0224 data. The first layer shows real bands, and the second layer shows spectral reconstruction bands.

which is the best of the five methods in RMSE, MRAE, and SAM.

### *D. Classification-Based Evaluation*

As an advanced application, classification can be used to measure the effect of spectral reconstruction. Classification performance comparisons of different methods in terms of OA, AA, and KAPPA, as well as the accuracy for each class on datasets 0531.0224 and 0506, are shown in Tables VIII–X.

In the sense of spectral reconstruction, the reconstructed HSIs' classification ability should be higher than that of the original MS images and lower than that of the real HSIs. The closer the classification ability of reconstructed HSIs is to that of real HSIs, the better the reconstruction algorithm's performance is.

From the classification results of dataset 0531 and dataset 0224, it can be seen that the HSIs reconstructed by PI have little difference in the classification ability compared with the original MS images. To some extent, it can be considered that the matrix-based mapping method cannot effectively transform MS images into high-quality HSIs. The classification effect of HSIs reconstructed by the PSJN and NPSJN methods proposed by us is optimal compared with the other four methods, whether OA, AA, or KAPPA. In addition, the reconstructed HSIs of PSJN and NPSJN have the optimal classification effect in several classes. In general, the classification performance of the reconstructed HSIs by PSJN is much higher than that of original MS images and close to that of real HSIs.

Since there is no corresponding HSI in dataset 0506, the algorithm's spectral reconstruction capability is measured by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

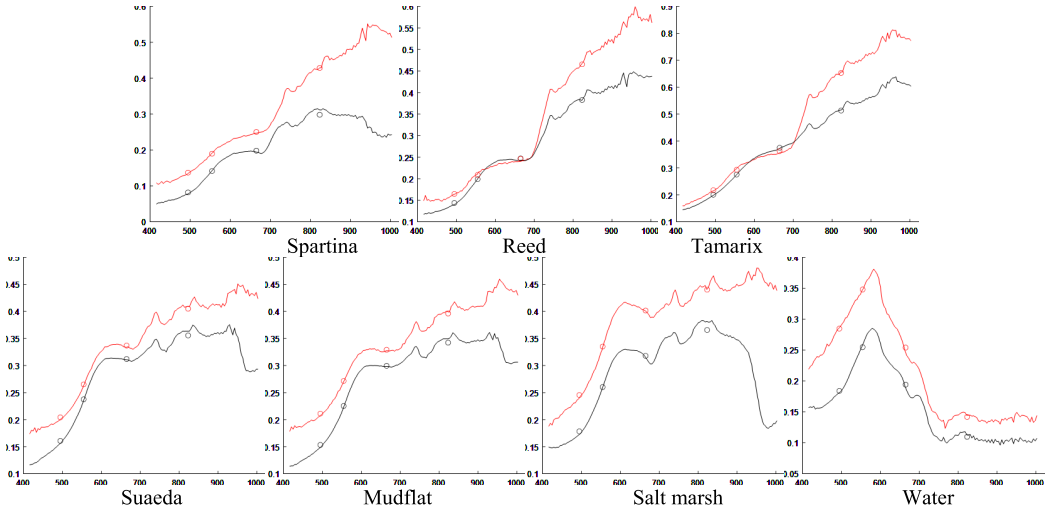



Fig. 15. Generated spectral reflectance curves for seven classes in dataset 0506 (black) and real spectral reflectance curves for seven classes in dataset 0531 (red).

TABLE VIII

CLASSIFICATION PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA (%), AA (%), AND KAPPA (%), AS WELL AS THE ACCURACY FOR EACH CLASS ON DATASET 0531. THE BEST RESULTS ARE SHOWN IN BOLD

| R0531 | PI | ARAD | J-SLoL | AWAN | NPSJN | PSJN | GF5 | GF1 |
|---|---|---|---|---|---|---|---|---|
| OA | 0.841 | 0.860 | 0.858 | 0.862 | **0.867** | **0.867** | 0.869 | 0.838 |
| AA | 0.837 | 0.840 | 0.851 | 0.845 | **0.855** | **0.854** | 0.854 | 0.831 |
| KAPPA | 0.796 | 0.817 | 0.817 | 0.820 | **0.825** | **0.828** | 0.831 | 0.791 |
| Spartina | 0.911 | 0.944 | **0.954** | 0.949 | 0.953 | **0.954** | 0.956 | 0.910 |
| Reed | **0.790** | 0.782 | 0.774 | 0.777 | 0.782 | 0.780 | 0.774 | 0.774 |
| Tamarix | 0.652 | 0.693 | 0.699 | 0.686 | **0.743** | **0.749** | 0.730 | 0.680 |
| Suaeda | 0.979 | 0.979 | 0.976 | **0.986** | 0.983 | 0.979 | 0.983 | 0.983 |
| mudflat | 0.647 | 0.574 | **0.661** | 0.600 | 0.609 | 0.600 | 0.612 | 0.614 |
| salt marsh | 0.978 | 0.966 | **0.989** | 0.978 | **0.989** | **0.989** | 0.989 | 0.955 |
| Water | 0.898 | **0.942** | 0.901 | 0.937 | 0.927 | 0.931 | 0.936 | 0.897 |

TABLE IX

CLASSIFICATION PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA (%), AA (%), AND KAPPA (%), AS WELL AS THE ACCURACY FOR EACH CLASS ON DATASET 0224. THE BEST RESULTS ARE SHOWN IN BOLD

| R0224 | PI | ARAD | J-SLoL | AWAN | NPSJN | PSJN | GF5 | GF1 |
|---|---|---|---|---|---|---|---|---|
| OA | 0.849 | 0.853 | 0.848 | 0.851 | **0.858** | **0.861** | 0.861 | 0.854 |
| AA | 0.836 | 0.835 | 0.836 | 0.841 | **0.849** | **0.842** | 0.855 | 0.835 |
| KAPPA | 0.826 | 0.825 | 0.824 | 0.825 | **0.830** | **0.829** | 0.834 | 0.821 |
| Water | **0.974** | **0.974** | **0.974** | **0.974** | **0.974** | 0.942 | 0.974 | 0.969 |
| Tree | 0.922 | 0.889 | 0.922 | 0.922 | **0.933** | 0.900 | 0.933 | 0.900 |
| Bare land | 0.769 | 0.769 | 0.769 | 0.788 | **0.808** | 0.769 | 0.846 | 0.769 |
| Field | 0.915 | 0.926 | 0.908 | 0.903 | **0.935** | **0.950** | 0.930 | 0.945 |
| Village | 0.839 | 0.849 | 0.841 | **0.866** | 0.844 | 0.834 | 0.856 | 0.819 |
| Residential | 0.670 | 0.672 | 0.665 | 0.663 | **0.680** | **0.682** | 0.685 | 0.690 |
| high-rise | 0.764 | 0.767 | **0.770** | **0.770** | 0.767 | **0.783** | 0.764 | 0.757 |

the degree of improvement in classification performance compared with that of the real MS image.

The OA of PI improved by 0.5%. The OA of ARAD and J-SLoL as dictionary methods is improved by 1.1% and 1.2%, respectively. The OA of AWAN is improved by 2.7%.

TABLE X

CLASSIFICATION PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF OA (%), AA (%), AND KAPPA (%), AS WELL AS THE ACCURACY FOR EACH CLASS ON DATASET 0506. THE BEST RESULTS ARE SHOWN IN BOLD

| R0506 | PI | ARAD | J-SLoL | AWAN | NPSJN | PSJN | GF1 |
|---|---|---|---|---|---|---|---|
| OA | 0.789 | 0.795 | 0.796 | 0.811 | **0.815** | **0.823** | 0.784 |
| AA | 0.747 | 0.732 | 0.738 | 0.758 | 0.756 | **0.760** | 0.740 |
| KAPPA | 0.733 | 0.740 | 0.740 | 0.760 | **0.764** | **0.773** | 0.729 |
| Spartina | 0.790 | 0.935 | 0.967 | 0.881 | **0.979** | **0.987** | 0.868 |
| Reed | 0.405 | 0.310 | 0.381 | **0.438** | 0.371 | 0.397 | 0.321 |
| Tamarix | 0.551 | **0.565** | 0.311 | 0.532 | 0.448 | 0.373 | 0.449 |
| Suaeda | 0.872 | 0.844 | 0.894 | 0.841 | **0.908** | 0.858 | 0.980 |
| mudflat | 0.750 | 0.629 | 0.753 | 0.759 | 0.742 | **0.834** | 0.716 |
| salt marsh | 0.859 | 0.839 | 0.861 | 0.857 | 0.85 | **0.869** | 0.846 |
| Water | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

The OA of the proposed PSJN method and simplified NPSJN method has been improved by 3.9% and 3.1%, respectively, while most classes' accuracies have been improved. It can be seen that the spectral reconstruction capability of our method is optimal among several methods.

Generated spectral reflectance curves for seven classes in dataset 0506 and real spectral reflectance curves for seven classes in dataset 0531 are showed in Fig. 15. The red and black circles represent the MS values of 0531 and 0506, respectively, and the red and black lines represent the HS spectral curves of 0531 and 0506, respectively. It can be seen that the spectral curves of real HSIs and generated HSIs are consistent in the same class.

We also classify the whole MS image and the reconstructed HSI by PSJN in dataset 0506, as shown in Fig. 16. The HS classification map of 0531 with near time is used as a reference for the analysis. It can be seen that, after spectral reconstruction, the misclassifications of mudflat and salt mash, spartina, and tamarix in the estuary have significantly been reduced. The division of the reeds in the middle of the river is also much clearer.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

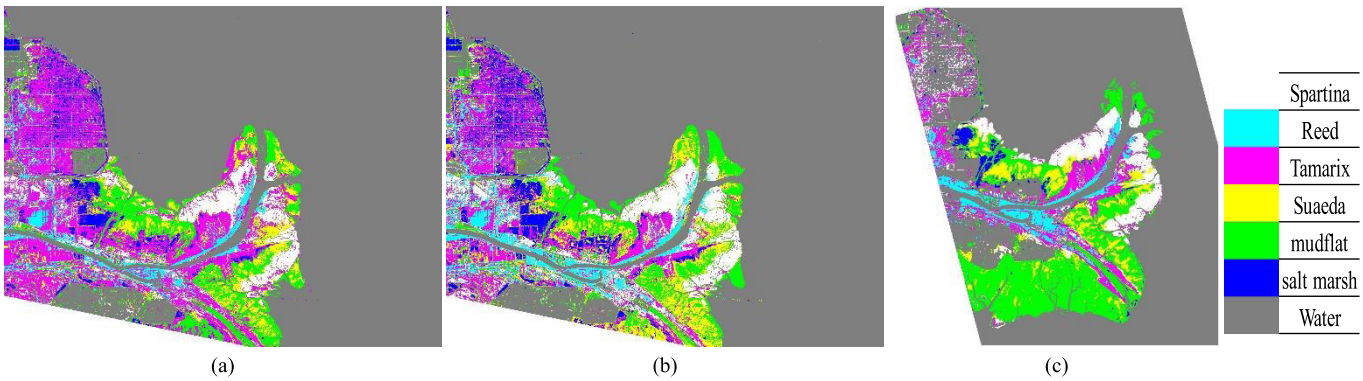



Fig. 16. Result of classification on dataset 0506. (a) MS image. (b) Reconstructed HSI by PSJN. (c) Dataset 0531 at the same place and near time as a reference.

TABLE XI
EFFECTS OF FIVE SITUATIONS. THE SAM METRIC IS MEASURED ON DATASETS 0531, 0417, AND 0224

| SAM | 0531 | 0417 | 0224 |
|---|---|---|---|
| PSJN | **0.01897** | **0.03175** | **0.05369** |
| PSJN-2D | 0.02051 | 0.03511 | 0.07090 |
| PSJN-3D | 0.01912 | 0.03301 | 0.05494 |
| PSJN-SE | 0.01903 | 0.03358 | 0.05433 |
| PSJN-2Dcat | 0.02018 | 0.03350 | 0.05922 |
| PSJN-3Dcat | 0.01928 | 0.03342 | 0.05660 |
| Ideal | ↓0 | ↓0 | ↓0 |

TABLE XII
EFFECTS OF DIFFERENT WEIGHT OF LOSS FUNCTIONS. THE SAM METRIC IS MEASURED ON DATASETS 0531, 0417, AND 0224

| $l_M$ / $l_H$ | 0531 | 0417 | 0224 |
|---|---|---|---|
| 1 | **0.01897** | **0.03175** | **0.05369** |
| 2 | 0.01920 | 0.03287 | 0.05610 |
| 5 | 0.01918 | 0.03314 | 0.05827 |
| 0.5 | 0.01900 | 0.03230 | 0.05395 |
| 0.2 | 0.01897 | 0.03301 | 0.05413 |
| 0 | 0.01902 | 0.03325 | 0.05423 |
| Ideal | ↓0 | ↓0 | ↓0 |

TABLE XIII
RECONSTRUCTION PERFORMANCE COMPARISON OF DIFFERENT METHODS IN TERMS OF THREE INDICES ON DATASET 0224

| | PI | ARAD | J-SLoL | AWAN | NPSJN | PSJN |
|---|---|---|---|---|---|---|
| Time/s | 0.5 | 33.9 | 3929 | 19.5 | 33.3 | 653 |

### *E. Network Ablation*

In order to determine the necessity of each part of the PSJN, a part of the network structure was removed for training, and the reconstruction results obtained were compared with those obtained from the original structure training.

There are a total of five situations for comparison: remove the 3-D part and only retain the 2-D part, remove the 2-D part and only retain the 3-D part, remove the SE part, remove the concatenation part in the 2-D part, and remove the concatenation part in the 3-D part.

Several situations' spectral reconstruction capability was compared by calculating the SAM between the reconstructed image and real image on datasets 0531, 0417, and 0224. The results are shown in Table XI.

It can be seen that the generation effect obtained by the complete structure of PSJN is better than in any situation where the partial structure is removed. The 2-D SPEM structure in PSJN can achieve a relatively good spectral reconstruction effect. The 3-D structure can enhance the correlation between bands but also increase the time spent on training and use. With the assistance of the 3-D PSFCM structure's spatial–spectral information, the full PSJN network with 2-D SPEM and 3-D PSFCM can obtain a better spectral reconstruction effect than 2-D SPEM itself.

The loss function of the PSJN network contains two parts: the HS-related loss ($l_H$) and the MS-related loss ($l_M$). Adjust the weights among loss to conduct the experiment, and the experimental results obtained are shown in Table XII. It can be seen that the addition of multispectral-related loss can improve the spectral reconstruction effect. However, when the weight ($l_M/l_H$) is too large, the reconstruction effect will be reduced. When the weight is 1, a relatively good reconstruction effect can be obtained.

### *F. Comparison of Computational Cost*

We also calculate the time required by different algorithms to reconstruct an MS image into an HSI. The reconstruction time of different methods in figure 0224 of four million pixels is shown in Table XIII. It can be seen that the PI method with the worst reconstruction effect has the fastest spectral reconstruction speed, which only takes 0.5 s. AWAN, NPSJN, and ARAD methods are the next fastest. PSJN is a bit slower, and J-SLoL is the slowest.

## V. CONCLUSION

In this article, we proposed a PSJN to reconstruct HSIs from MS images. PSJN consists of three parts: the 2-D spatial feature extraction module, the 3-D progressive spatial–spectral feature construction module, and the spectral postprocessing module. The network combines the shallow spatial features

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

extracted by the 2-D spatial feature extraction module with the spatial–spectral features extracted by the 3-D progressive spatial–spectral feature construction module and obtains the reconstructed HSIs through the spectral postprocessing module. The 3-D progressive module is designed to extract spatial–spectral information from local spectra in local space and construct spectral information from low frequency to high frequency in a pyramidal structure. A network updating mechanism is set up to deal with the situation that the data distribution of the MS images to be reconstructed is too different from that of the MS images during training to improve the spectral reconstruction effect. Extensive experiments on three HS–MS datasets and one MS dataset to evaluate the similarity between the reconstructed HSIs and the real HSIs and the classification ability of reconstructed HSIs demonstrate the effectiveness of our proposed PSJN. Although the proposed PSJN has already had excellent spectral reconstruction capability, it can be seen that there is still room for progress in the spectral reconstruction across scenes and classes. In future work, we would like to develop SR approaches on multisensor and multiscene information to further improve the models' spectral reconstruction capability.

## ACKNOWLEDGMENT

The authors would like to thank the First Institute of Oceanography of Natural Resources for providing the raw data of the GF5 dataset. They would also like to thank B. Arad, D. Hong, and J. Li for providing the code of ARAD, J-SLoL, and the adaptive weighted attention network (AWAN).

## REFERENCES

[1] Y. Gu, T. Liu, and J. Li, "Superpixel tensor model for spatial–spectral classification of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4705–4719, Jul. 2019.

[2] Y. Chen, Y. Wang, Y. Gu, X. He, P. Ghamisi, and X. Jia, "Deep learning ensemble for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1882–1897, Jun. 2019.

[3] Y. Niu and B. Wang, "Extracting target spectrum for hyperspectral target detection: An adaptive weighted learning method using a self-completed background dictionary," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1604–1617, Mar. 2017.

[4] X. Jin, Y. Gu, and T. Liu, "Intrinsic image recovery from remote sensing hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 224–238, Jan. 2019.

[5] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2019.

[6] S. Zhang, J. Li, H. Li, C. Deng, and A. Plaza, "Spectral–spatial weighted sparse regression for hyperspectral image unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3265–3276, Jun. 2018.

[7] Y. Tarabalka, J. Chanussot, and J. A. Benediktsson, "Segmentation and classification of hyperspectral images using watershed transformation," *Pattern Recognit.*, vol. 43, no. 7, pp. 2367–2379, Jul. 2010.

[8] T. Liu, X. Zhang, and Y. Gu, "Unsupervised cross-temporal classification of hyperspectral images with multiple geodesic flow kernel learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9688–9701, Dec. 2019.

[9] Y. Gu *et al.*, "Multimodal hyperspectral remote sensing: An overview and perspective," *Sci. China Inf. Sci.*, vol. 64, no. 2, Feb. 2021, Art. no. 121301.

[10] K. Zhang, M. Wang, and S. Yang, "Multispectral and hyperspectral image fusion based on group spectral embedding and low-rank factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1363–1371, Mar. 2017.

[11] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[12] R. Guerra, S. Lopez, and R. Sarmiento, "A computationally efficient algorithm for fusing multispectral and hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 5712–5728, Oct. 2016.

[13] Y. F. Gu *et al.*, "UAV-based integrated multispectral-LiDAR imaging system and data processing," *Sci. China Technol. Sci.*, vol. 63, no. 7, pp. 1293–1301, 2020.

[14] X. Zhang and H. Xu, "Reconstructing spectral reflectance by dividing spectral space and extending the principal components in principal component analysis," *J. Opt. Soc. Amer. A, Opt. Image Sci.*, vol. 25, no. 2, pp. 371–378, 2008.

[15] X. Liu and L. Liu, "Improving chlorophyll fluorescence retrieval using reflectance reconstruction based on principal components analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 8, pp. 1645–1649, Aug. 2015.

[16] H. Haneishi, T. Hasegawa, A. Hosoi, Y. Yokoyama, N. Tsumura, and Y. Miyake, "System design for accurately estimating the spectral reflectance of art paintings," *Appl. Opt.*, vol. 39, no. 35, pp. 6621–6632, Dec. 2000.

[17] F. H. Imai and R. S. Berns, "Spectral estimation using trichromatic digital cameras," in *Proc. Int. Symp. Multispectral Imag. Color Reproduction Digital Archives*, 1999, pp. 42–49.

[18] V. Cheung, S. Westland, C. Li, J. Hardeberg, and D. Connah, "Characterization of trichromatic color cameras by using a new multispectral imaging technique," *J. Opt. Soc. Amer. A, Opt. Image Sci., Vis.*, vol. 22, no. 7, pp. 1231–1240, Jul. 2005.

[19] N. T. Hoang and K. Koike, "Transformation of landsat imagery into pseudo-hyperspectral imagery by a multiple regression-based model with application to metal deposit-related minerals mapping," *ISPRS J. Photogramm. Remote Sens.*, vol. 133, pp. 157–173, Nov. 2017.

[20] B. Arad and O. Ben-Shahar, "Sparse recovery of hyperspectral signal from natural RGB images," in *Proc. ECCV*, 2016, pp. 19–34.

[21] A. Robles-Kelly, "Single image spectral reconstruction for multimedia applications," in *Proc. 23rd ACM Int. Conf. Multimedia*, Oct. 2015, pp. 251–260.

[22] J. Wu, J. Aeschbacher, and R. Timofte, "In defense of shallow learned spectral reconstruction from RGB images," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Venice, Italy, Oct. 2017, pp. 471–479.

[23] Y. Jia *et al.*, "From RGB to spectrum for natural scenes via manifold-based mapping," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4715–4723.

[24] X. Han, J. Yu, J. Luo, and W. Sun, "Reconstruction from multispectral to hyperspectral image using spectral library-based dictionary learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1325–1335, Mar. 2019.

[25] C. Yi, Y.-Q. Zhao, and J. C.-W. Chan, "Spectral super-resolution for multispectral image based on spectral improvement strategy and spatial preservation strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9010–9024, Nov. 2019.

[26] K. Fotiadou, G. Tsagkatakis, and P. Tsakalides, "Spectral super resolution of hyperspectral images via coupled dictionary learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2777–2797, May 2019.

[27] L. Gao, D. Hong, J. Yao, B. Zhang, P. Gamba, and J. Chanussot, "Spectral superresolution of multispectral imagery with joint sparse and low-rank learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2269–2280, Mar. 2021.

[28] R. M. Nguyen, D. K. Prasad, and M. S. Brown, "Training-based spectral reconstruction from a single RGB image," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 186–201.

[29] S. Galliani, C. Lanaras, D. Marmanis, E. Baltsavias, and K. Schindler, "Learned spectral super-resolution," 2017, *arXiv:1703.09470*. [Online]. Available: http://arxiv.org/abs/1703.09470

[30] Z. Xiong, Z. Shi, H. Li, L. Wang, D. Liu, and F. Wu, "HSCNN: CNN-based hyperspectral image recovery from spectrally undersampled projections," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Venice, Italy, Oct. 2017, pp. 518–525.

[31] A. Alvarez-Gila, J. Van De Weijer, and E. Garrote, "Adversarial networks for spatial context-aware spectral image reconstruction from RGB," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, Venice, Italy, Oct. 2017, pp. 480–490.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.

[32] S. Paul and D. N. Kumar, "Transformation of multispectral data to quasi-hyperspectral data using convolutional neural network regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3352–3368, Apr. 2021, doi: 10.1109/TGRS.2020.3009290.

[33] Z. Shi, C. Chen, Z. Xiong, D. Liu, and F. Wu, "HSCNN+: Advanced CNN-based hyperspectral recovery from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Salt Lake City, UT, USA, Jun. 2018, pp. 1052–10528.

[34] J. Li, C. Wu, R. Song, Y. Li, and F. Liu, "Adaptive weighted attention network with camera spectral sensitivity prior for spectral reconstruction from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Seattle, WA, USA, Jun. 2020, pp. 1894–1903.

[35] Y. Zhao, L.-M. Po, Q. Yan, W. Liu, and T. Lin, "Hierarchical regression network for spectral reconstruction from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Seattle, WA, USA, Jun. 2020, pp. 1695–1704.

[36] Y.-T. Lin and G. D. Finlayson, "Physically plausible spectral reconstruction from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, JSeattle, WA, USA, un. 2020, pp. 2257–2266.

[37] J. Li *et al.*, "Hybrid 2-D–3-D deep residual attentional network with structure tensor constraints for spectral super-resolution of RGB images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2321–2335, Mar. 2021.

[38] C.-C. Hsu, C.-H. Lin, C.-H. Kao, and Y.-C. Lin, "DCSN: Deep compressed sensing network for efficient hyperspectral data transmission of miniaturized satellite," *IEEE Trans. Geosci. Remote Sens.*, pp. 1–17, 2020.

[39] B. Arad *et al.*, "NTIRE 2018 challenge on spectral reconstruction from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Salt Lake City, UT, USA, Jun. 2018, pp. 1042–104209.

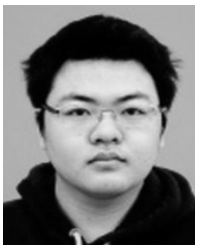
[40] B. Arad *et al.*, "NTIRE 2020 challenge on spectral reconstruction from an RGB image," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Seattle, WA, USA, Jun. 2020, pp. 1806–1822.

[41] H. Zhao, G. Jia, and N. Li, "Transformation from hyperspectral radiance data to data of other sensors based on spectral superresolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 3903–3912, Nov. 2010.

[42] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.

[43] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Santiago, Chile, Dec. 2015, pp. 1026–1034.

[44] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 7132–7141.



**Tianshuai Li** received the B.E. degree in electronics and information engineering from the Harbin Institute of Technology, Harbin, China, in 2018, where he is pursuing the Ph.D. degree in information and communication engineering.

His research interests include deep learning and its application to remote sensing data analysis, and hyperspectral image processing.



**Yanfeng Gu** (Senior Member, IEEE) received the Ph.D. degree in information and communication engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2005.

He joined the School of Electronics and Information Engineering, HIT, as a Lecturer, where he was appointed as an Associate Professor in 2006. From 2011 to 2012, he was a Visiting Scholar with the Department of Electrical Engineering and Computer Science, University of California at Berkeley, Berkeley, CA, USA. He is a Professor with the Department of Information Engineering, HIT. He has authored more than 100 peer-reviewed articles and 4 book chapters. He holds 20 patents. His research interests include hyperspectral remote sensing image processing, intelligent satellite data processing, and unmanned aerial vehicle (UAV)-based hyperspectral LiDAR detection and processing.

Dr. Gu serves as an Associate Editor for the IEEE Transactions on Geoscience and Remote Sensing and *Science China Technological Sciences*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:42:33 UTC from IEEE Xplore. Restrictions apply.